

$$\begin{array}{c} \mathbf{Q} \\ \text{ } \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array} \end{array} \cdot \begin{array}{c} \mathbf{K}^T \\ \text{ } \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array} \end{array} = \begin{array}{|c|c|c|} \hline x_{11} & x_{12} & x_{13} \\ \hline x_{21} & x_{22} & x_{23} \\ \hline x_{31} & x_{32} & x_{33} \\ \hline \end{array}$$

$\frac{dx}{dt} = \frac{dA}{dt}$

High
Variance



Vanishing
gradient Problem



Hinders
Training

$\text{Variance}(A \cdot B) \propto n$, where n = common dimension of A and B

$$\begin{array}{c} \mathbf{Q} \end{array} \cdot \begin{array}{c} \mathbf{K}^T \end{array} = \begin{array}{|c|c|c|} \hline x_{11} & x_{12} & x_{13} \\ \hline x_{21} & x_{22} & x_{23} \\ \hline x_{31} & x_{32} & x_{33} \\ \hline \end{array}$$

$d_k = 64$

High
Variance



Vanishing
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1. Why high variance?

2. Why Variance leads to VGP?

3. Why divide by $\sqrt{d_k}$

$\text{Variance}(A \cdot B) \propto n,$ where n = common dimension of A and B

$$\begin{aligned} & \text{(-10, 10)} \\ & \downarrow \\ & [10, 10] \times [10, 10] \\ & = 2 \times 100 \end{aligned}$$

$$\underline{[4, 0] \times [5, -9] = 20}$$

$$\underline{[-3, -8] \times [-3, 4] = -23}$$

$$\underline{[1, -5] \times [-1, 4] = -21}$$

$$\underline{[-4, 8] \times [0, 0] = 0}$$

$$\underline{[10, 7] \times [-4, 9] = 23}$$



$$\begin{aligned} & [10, 10, 10, 10, 10] \times [10, 10, 10, 10, 10] \\ & = 5 \times 100 \end{aligned}$$

$$\underline{[-7, 7, -3, -7, -9] \times [-4, 10, -2, -4, 7] = 69}$$

$$\underline{[-3, 4, -8, 3, 6] \times [-9, -1, -7, 3, 5] = 118}$$

$$\underline{[-7, 3, 7, -2, 10] \times [-1, 5, 4, 4, 8] = 122}$$

$$\underline{[4, -3, 3, -3, 10] \times [-5, -1, -7, 7, 1] = -49}$$

$$\underline{[-9, 9, 4, -4, 1] \times [1, 9, -8, -6, 8] = 72}$$



Variance($A \cdot B$) $\propto n$, where n = common dimension of A and B

$$C = A \cdot B$$

$$\text{Var}(C) = n \cdot \text{Var}(\underline{A}) \cdot \text{Var}(\underline{B})$$

- Independent random variables
- Zero mean
- Entries of A and B have identical variance

$$\text{Var}(C) \approx n \times \text{Var}(A) \cdot \text{Var}(B)$$

Variance($A \cdot B$) $\propto$ n , where n = common dimension of A and B

$$X = Q \cdot K^T$$

$$\text{Var}(X) \propto d_k$$

$$\begin{aligned} \underline{d_k = 1}, \quad \text{Var}(X) &= A \rightarrow A \\ \underline{d_k = 2}, \quad \text{Var}(X) &= 2 \cdot A \rightarrow A \\ \underline{d_k = 3}, \quad \text{Var}(X) &= 3 \cdot A \rightarrow A \\ &\vdots \\ \underline{d_k = 64}, \quad \text{Var}(X) &= 64 \cdot A \rightarrow A \end{aligned}$$

$$a = \frac{1}{\sqrt{d_k}}$$

$$\begin{aligned} X &\rightarrow \text{Var}(X) \\ a \cdot X &\rightarrow a^2 \cdot \text{Var}(X) \end{aligned}$$

$$d_k = 1, \quad \text{Var}(X) = A$$

$$\begin{aligned} d_k = d_k, \quad \text{Var}(X) &= d_k \cdot A \\ &= d_k \cdot \left(\frac{1}{\sqrt{d_k}}\right)^2 \cdot A \\ &= A \end{aligned}$$

$$\text{Attention} = \text{Softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) \cdot V$$